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Improved soot prediction models for turbulent non-premixed ethylene/air flames

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Abstract

Existing soot models for non-premixed ethylene/air flames often do not satisfactorily predict soot volume fraction. Due to intense energy exchange by radiative heat transfer, this leads to a situation where even basic scalar fields like temperature are inaccurately described. The objective of the present study is to develop an enhanced soot prediction model for such flames, valid for a wide range of flow and operating conditions. The emphasis in this work is specifically on the particulate phase and not on kinetics. Therefore, gas-phase chemistry is simply modeled with a mixture fraction approach combined with equilibrium chemistry, while turbulence–chemistry interactions are taken into account by a presumed β-PDF. The evolution of soot particles is described by physical models accounting for nucleation, surface growth, aggregation and oxidation. The Direct Quadrature Method of Moments (DQMOM) is employed to solve the Population Balance Equations (PBE) in a computationally efficient manner, assuming a mono-variate PBE with particle diameter as internal coordinate. The original soot model is optimized numerically by using Genetic Algorithms coupled with Computational Fluid Dynamics (CFD). The values of three model parameters, associated with nucleation, oxidation and soot particle radiation are optimized through comparison with recent experimental results. Two objective functions are formulated based on the difference between experimental and simulation results for temperature and soot volume fraction. After obtaining an optimal parameter set, the resulting model is further tested against three experimental configurations, leading to a good agreement and thus demonstrating a high level of generality.
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Keywords: Soot volume fraction; Non-premixed flames; Direct quadrature method of moments; Optimization; Mono-variate PBE

1. Introduction

Due to the adverse effects of soot on human health as a carcinogenic [1,2] and on the environment as one major global warming component [3,4] with a high impact on cloud cover and visibility [5], there have been numerous studies during the last decades to understand soot formation and its evolution. Nevertheless, many questions remain unsolved due to the tremendous complexity of this process, involving gas-phase and surface kinetics, coupling with turbulence, large heat transfer by radiation and intricate interactions between the particles. Soot models are often

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Nomenclature

C_0	radiation proportionality factor, m^{-1} K^{-1}
D_i	internal nozzle diameter, mm
E	error for soot volume fraction
e	error for temperature
f_v	soot volume fraction
G_g, G_o	molecular & oxidation growth rate, ms^{-1}
ΔH	relative difference in peak values
J	nucleation rate, m^{-3} s^{-1}
$\dot{m}$	mass flow rate, g/s
$OF1$	first objective function
$OF2$	second objective function
T	temperature, K
u_i	injection velocity, m/s

Greek symbols

β_{12}	aggregation kernel, m^3s^{-1}

Subscripts

g	molecular growth
n	nucleation
o	oxidation
s	soot

classified into three different groups [6]: (i) empirical, (ii) semi-empirical and (iii) kinetic soot models. While empirical models do not allow detailed predictions, kinetic soot models are obviously best suited to describe all the complex pathways leading from the fuel to the final particles. A detailed modeling of soot formation based on polycyclic aromatic hydrocarbon (PAH) concentration is given for instance in [7]. Nevertheless, such models are difficult to employ for real, three-dimensional flames of practical interest, due to the considerable computational time required by the simulations [8]. As a consequence, semi-empirical models are mostly used at present and are therefore the target of this work. They involve less computational effort by simplifying all kinetic aspects. Usually, one single chemical species is deemed responsible for soot nucleation and rate equations are developed based only on this species. For example, acetylene [9,10], benzene, naphthalene or PAH [11] are considered to be responsible for soot inception in ethylene/air flames. In this manner, the key processes controlling soot formation and evolution of the solid phase are then nucleation, surface growth, aggregation and oxidation [12,13]. They must be described by suitable physical models, together with radiative heat transfer. As in the present work, molecular growth of the soot particles is mostly associated with acetylene in ethylene/air flames, while oxidation rates are usually computed from local oxygen concentration.

In the past, several authors tried to propose suitable combinations of particle phase models in order to predict soot production. For instance, Ma et al. tried different combinations of inception, growth, oxidation and aggregation models in order to determine the best parameters for a non-premixed ethylene/air flame [11]. Zucca et al. proposed another model combination for the same type of flames, again validated against experimental results [14]. Being more recent, the model formulation presented in [14] has been retained as initial solution for the optimization process described in what follows. Even if such studies have led to a significant improvement in understanding and predicting soot formation, available models have yet to yield satisfactory comparisons against all available experimental data [15,16]. There is thus a clear need for more accurate prediction models, applicable to a wide range of flow and operating conditions. First computational studies in our group concerning predictions of soot volume fraction [17] have demonstrated that the nucleation and oxidation models play a first-order role for typical flame configurations, followed by molecular growth and finally aggregation processes. Therefore, nucleation and oxidation parameters are first retained in the present optimization.

Due to its high absorption coefficient, soot formation is closely associated with radiative heat loss and thus influences the temperature field, leading to an indirect coupling with flame structure and chemical pathways [6]. Inadequate treatment of radiation terms leads to very large errors. An analysis of existing soot radiation models (e.g., [18,19]) shows that there is still a high level of uncertainty concerning the most appropriate model description. As a consequence, this aspect is taken as well into account in the optimization process.

The present numerical optimization relies on Computational Fluid Dynamics (CFD) to find the best possible model parameters. CFD-based optimization (CFD-O [20]) has been successfully applied to different configurations involving heat transfer and flames within our research group [21,22]. Based on our past experience, a Genetic Algorithm (GA) coded in the in-house optimization library OPAL (OPtimization ALgorithms) has been combined with the industrial CFD solver ANSYS-Fluent for the present study. The particulate phase must be solved in a coupled manner together with the reactive Navier–Stokes equations. For this purpose, Frenklach and co-workers

applied Method of moments (MOM) for modeling soot formation in laminar premixed flames [23,24]. In order to obtain a computationally efficient solution, Direct Quadrature Method of Moments (DQMOM) has been coded for the present study into ANSYS-Fluent using User-Defined Functions. With DQMOM, the number of additional transport equations needed to describe the particle phase is relatively small (typically 4–6 [25]), thus speeding up the computations compared to a direct solution of the PBE [26].

Four different turbulent non-premixed ethylene/air flames [27–30] have been considered in this work. The first flame [27] is employed to determine the optimal set of model parameters by comparing the obtained fields of temperature and soot volume fraction. The optimal model is then checked by comparison with the three other flames [28–30], showing an improved agreement compared to the original model. After describing the considered configuration in what follows, the numerical models will be introduced. Then, the optimization strategy will be briefly presented, before discussing the results.

2. Configuration

All turbulent non-premixed ethylene/air flames considered in this work have a similar geometry, as shown in Fig. 1(a). Pure ethylene enters into quiescent air through a nozzle and combustion takes place at atmospheric pressure (open configuration). As the domain is symmetrical with respect to the central axis, simulations are carried out in an axisymmetric configuration (Fig. 1(b)). Internal nozzle diameter (D_i in mm), fuel mass flow rate ($\dot{m}$ in g/s), average exit velocity (u_i in m/s) and associated Reynolds number (Re) of the four different flames considered in the present study are listed in Table 1. All flames are turbulent.

Grid generation has been carried out using the software Gambit with roughly 20,000 grid points, used to discretize an axisymmetric computational domain of 300 mm (width) × 900 mm (length), kept identical for all flames. A non-uniform mesh is employed for all computations, with very fine cells near the axis and coarser cells near domain boundary. The size of the domain and the computational grid have been retained after a systematic study of the dependence of the results on the grid resolution and on the influence of outflow boundary conditions. In this manner, it has been verified that the obtained fields only depend on the employed physical models.

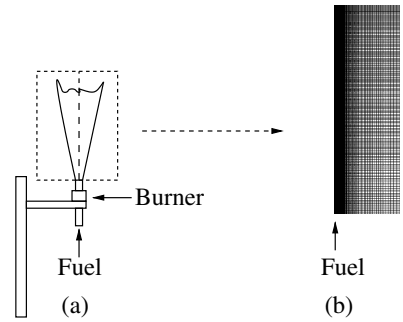

Fig. 1. General configuration: (a) sketch of experimental set-up, (b) associated adapted computational grid.

A multi-objective optimization is carried out by considering two objective functions and three free model parameters. The two objective functions are the differences between experimental data and simulation results concerning the fields of temperature and of soot volume fraction. The three free parameters have been selected after a sensitivity analysis of each model parameter appearing in the original formulation [14]. Optimization is based on the experimental results of [27], while three additional flame types are used to further ascertain the generality of the optimized solution.

3. Employed models

3.1. Computational fluid dynamics

The industrial CFD software ANSYS-Fluent is employed in the present study to solve all gas-phase transport equations. The Favre-averaged Navier–Stokes equations are solved for this purpose. Turbulence closure is obtained through the standard $k-\epsilon$ turbulence model, suitable for this simple flow configuration (axisymmetric jet).

3.2. Gas-phase chemistry

As already explained previously, the emphasis here is not set on the extremely difficult issue of gas-phase or surface kinetics. Therefore, gas-phase chemistry is described simply through a mixture

Table 1
Experimental conditions considered in the simulations.

Flame	D_i (mm)	$\dot{m}$ (g/s)	u_i (m/s)	Re	Reference
I	4.6	0.48	25.4	13,500	[27]
II	4.56	0.48	25.8	13,500	[28]
III	3.0	0.419	52.2	14,660	[29]
IV	2.18	0.209	48.7	12,000	[30]

fraction approach with equilibrium chemistry approximation. Turbulence/chemistry interactions are modeled through a presumed β-PDF approach, solving the transport equation for mixture fraction $\widetilde{Z}$ and for its variance $\widetilde{Z''^2}$ [31]. Twenty chemical species (C_2H_4, O_2, H_2, CO, CH_4, CO_2, H_2O, C_2H_2, OH, H, O, C_6H_6, C_4H_2, C_2N_2, C_4H, C_3H_2, H_2CCCCH, C_2H_6, C_2N, N_2) are taken into account in the equilibrium calculation.

3.3. Particle phase models

As explained previously, the model combination proposed and validated in [14] has been retained as an original model for the optimization. As a consequence, the nucleation model introduced by [32] and the molecular growth model presented by [33] have been retained, both acetylene-based. Oxidation results only from O_2 molecules, as proposed in [34]. The rates of nucleation J, molecular growth G_g and oxidation G_o are finally written:

$$J = N_A \rho^2 T^{\frac{1}{2}} 6\ 10^6 \exp\left(-\frac{T_n}{T}\right) X_{C_2H_2} \tag{1}$$

$$G_g = \frac{6}{D_f \rho_s}\left(\frac{R_c}{R_{c0}}\right)^{\frac{3-D_f}{3}} 2M_s\ 6 \exp\left(-\frac{T_g}{T}\right)[C_2H_2] \tag{2}$$

$$G_o = \frac{p}{D_f \rho_s} T^{-\frac{1}{2}} 6.5 \exp\left(-\frac{T_o}{T}\right) Y_{O_2} \tag{3}$$

Here, X is mole fraction, Y is mass fraction, $M_s = 12$ g/mol is soot molecular weight, N_A is Avogadro's number, $[C_2H_2]$ represents the acetylene concentration. The parameters T_n, T_g and T_o are equivalent activation temperatures for nucleation, growth and oxidation, respectively.

The aggregation kernel β_{12}, which depends on the Knudsen number, $\mathrm{Kn} = \frac{2\lambda}{L}$ (the ratio between the molecular mean free path of gas molecules, λ and the particle radius, $L/2$) is evaluated by the Fuchs interpolation formula [14]:

$$\beta_{12} = 4\pi(D_1 + D_2)(R_{c1} + R_{c2})\left[\frac{R_{c1} + R_{c2}}{R_{c1} + R_{c2} + (g_1^2 + g_2^2)^{\frac{1}{2}}} + \frac{4(D_1 + D_2)}{R_{c1} + R_{c2} + (c_1^2 + c_2^2)^{\frac{1}{2}}}\right]^{-1} \tag{4}$$

where $c_i = \sqrt{\frac{8k_bT}{\pi m_i}}$, $D_i = \frac{k_bT}{6\pi\mu R_{ci}}\left[\frac{5+4\mathrm{Kn}_i+6\mathrm{Kn}_i^2+18\mathrm{Kn}_i^3}{5-\mathrm{Kn}_i+(8+\pi)\mathrm{Kn}_i^2}\right]$, $l_i = \frac{8D_i}{\pi c_i}$, $g_i = \frac{(2R_{ci}+l_i)^3-(4R_{ci}^2+l_i^2)^{\frac{3}{2}}}{6R_{ci}l_i} - 2R_{ci}$, with m the particle mass and k_b the Boltzmann constant. The parameter R_c is the collision radius, which is defined in terms of the fractal dimension D_f of the aggregate through

$$R_c = k_f \frac{L_0}{2}\left(\frac{V}{V_0}\right)^{\frac{1}{D_f}} \tag{5}$$

with V the aggregate volume. The subscript 0 indicates primary particles, assumed of size 15 nm, and k_f is the fractal prefactor of order unity. The present simulation results are computed by considering a constant fractal dimension $D_f = 2.0$ and $k_f = 1$, since a previous sensitivity analysis revealed that the predicted soot volume fraction is not very sensitive against D_f.

3.4. Direct quadrature method of moments (DQMOM)

The evolution of the particle phase is handled by Direct Quadrature Method of Moments. Moment-based methods have been proven to be accurate for predicting the moments of a variety of particle size distributions (PSD). Additionally, they require less computational time compared with a direct solution of the PBE or with sectional methods. DQMOM appears very attractive since optimization can only be carried out when each individual simulation is relatively fast (at most a few hours of computing time). The quadrature approximation [35] used in DQMOM solves the closure problem associated with nonlinearities in source terms such as growth and aggregation. With DQMOM, the PBE simplifies into a set of algebraic equations with few additional transport equations. For example, if we consider two nodes for the quadrature approximation as with the present case, four transport equations must be solved. For the sake of brevity the details about DQMOM are not given here and we refer the reader to [25] for a complete mathematical description.

3.5. Radiative heat transfer

The radiative heat transfer rate from the system Q_{rad} is highly significant for the problem considered here, since combustion occurs at high temperature and since soot particles have a large absorption coefficient. The radiation model implemented in [14] is retained as an original model to describe radiative heat transfer associated with the soot particles:

$$Q_{rad,s} = -\sigma a_s\left(T^4 - T_\infty^4\right) \tag{6}$$

where σ is the Stefan–Boltzmann constant, T_∞ is the surrounding environment temperature (here 300 K) and a_s is the soot absorption coefficient, given by:

$$a_s = C_0 f_v T \tag{7}$$

In this last equation, f_v is the soot volume fraction and $C_0 = 2370\ \mathrm{m^{-1}\ K^{-1}}$. This constant has been determined for smoke produced from organic fuels [19].

In addition, the radiative heat transfer resulting from the three gaseous species C_2H_4, CO_2 and H_2O has also been taken into account, con-

sidering mean absorption coefficients as given in [36]. The contributions from all other gaseous species are considered negligible.

4. Numerical optimization strategy

Four numerical tools are coupled for the present computations, as shown in Fig. 2. CFD is used to solve flow, energy and species transport for the gaseous phase, coupled through User-Defined Functions with DQMOM to obtain simultaneously the properties of the soot particle phase. After obtaining a converged flame solution, MATLAB is used to evaluate the two objective functions, comparing simulation results with published experimental data in an automatic post-processing step. The optimization algorithm (GA) then manipulates the free parameters in an attempt to further minimize the objective functions. A subroutine coded in C manages automatically all the above computational tools in the right sequence.

The axial profiles of temperature and soot volume fraction for the experiment of [27] are shown in Figs. 3 and 4, respectively. Circles are experimental data while lines are simulation results obtained with the original model of [14] and with the optimized model.

4.1. Objective function-1 (OF1)

The square symbols (□) in Fig. 3 are simulation data points corresponding to the locations where experimental data points (circles) are available. The differences between experimental and simulation results are exemplified in Fig. 3 with e_i, $e_{i+1}\ldots$ By adding up all such absolute errors for temperature, one obtains the first objective function, $OF1$, which should be minimized:

$$OF1 = |e_1| + |e_2| + |e_3| + |e_4| + \ldots \quad (8)$$

4.2. Objective function-2 (OF2)

In Fig. 4 soot volume fractions (f_v) are shown using experimental data obtained from two different measurement methods. Due to the considerable differences observed from the two measurements, average experimental results have been determined by a best fit of all experimental data points using a 5th order polynomial fit, represented by square symbols. Each simulation result is compared with the fitted experimental curve and absolute errors are again added as exemplified in Fig. 4 with E_i, $E_{i+1}\ldots$ This absolute error for soot volume fraction constitutes the second objective function, $OF2$, which should be minimized as well:

$$OF2 = |E_1| + |E_2| + |E_3| + |E_4| + \ldots \quad (9)$$

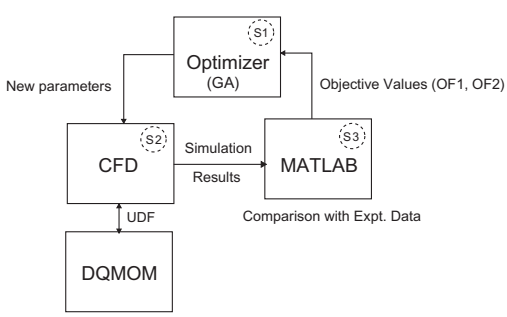

Fig. 2. Numerical tools employed for the optimization, with calling sequence shown with S1, S2 and S3.

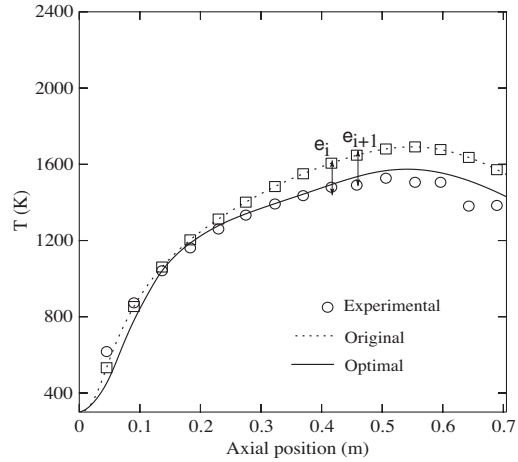

Fig. 3. Comparison of experimental temperature along flame axis with original and optimal soot model for Flame I.

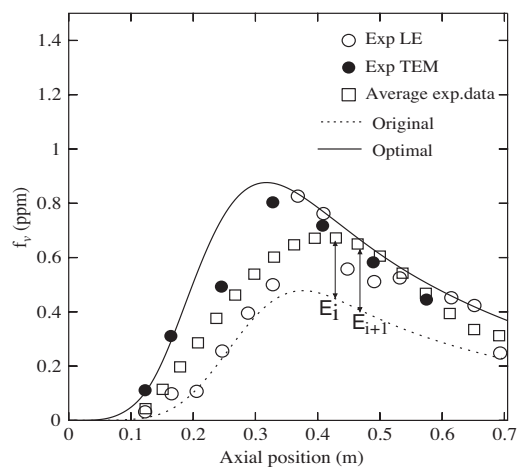

Fig. 4. Comparison of experimental soot volume fraction f_v along flame axis with original and optimal soot model for Flame I.

4.3. Experimental uncertainties

Since our research group carries out extensive experimental measurements in flames, we are well aware that each data point is associated with some level of experimental uncertainty, in particular for such complex optical techniques. Even if the level of uncertainty is not always explicitly stated in the literature, it should be taken into account in order to avoid an "excessive" optimization, since the procedure will do its best by default to put the simulation results on top of the experimental points. To take into account a realistic level of measurement uncertainty, the experimental data have been associated with a tolerance level of ±10% for temperature and of ±20% for soot volume fraction. These ranges are employed to compute penalty points. All simulations leading to results within this range of uncertainty are associated with a zero penalty and are therefore considered equally optimal. On the other hand, simulations falling outside this range become one penalty point for each corresponding position.

4.4. Optimization parameters

A previous sensitivity analysis performed on all the parameters of the soot particle model (Eqs. (1)–(5)) and radiation model (Eqs. (6) and (7)) has demonstrated that nucleation, oxidation and radiation parameters dominate the prediction of soot volume fraction. Therefore, three optimization parameters have been retained in the present study: the activation temperature for nucleation (T_n), for oxidation (T_o) and the soot radiation constant (C_0). After some manual tests, the variation range for the three parameters has been prescribed as $42,000 \leqslant T_n \leqslant 46,100$ [K], $24,000 \leqslant T_o \leqslant 26,500$ [K], $2400 \leqslant C_0 \leqslant 5000$ [m^{-1} K^{-1}].

The final goal of the optimization is to find the optimal set of parameters T_n, T_o and C_0 within the given parameter space, in such a way that the two objective functions $OF1$ and $OF2$ are simultaneously minimized without any penalty.

4.5. Computational details

The optimization sequence has been shown in Fig. 2:

Step 1, S1: the Optimization algorithm (GA) generates the new individuals (i.e., parameter sets) within the given parameter space, from the previous population.

Step 2, S2: Using these parameters CFD/DQMOM simulations are carried out in order to evaluate the resulting fields.

Step 3, S3: Converged CFD/DQMOM results are analyzed by comparison with experimental data and the objective functions $OF1$, $OF2$ (Eqs. 8-9) are computed using MATLAB.

Loop: These results are transferred back to the GA in order to generate the next generation.

CFD-O is performed completely automatically in batch mode using six computing cores in parallel. In this manner, assessing the properties of one individual takes roughly 4 h of computing time. As a whole, more than 60 different parameter sets have been tested by the optimization procedure. The optimization process has been able to identify a considerably better set of parameters compared to the original model, as documented in Figs.3 and 4. Both temperature and f_v were under predicted with the original model parameters. The predictions are improved significantly with the optimal set. The temperature profile becomes in very close agreement with the experimental data (Fig. 3). The peak soot volume fraction is much closer to the peak experimental value (≈0.8 ppm), while it was considerably lower with the original model (≈0.5 ppm). The profile of f_v is nevertheless slightly shifted towards the burner, a tendency observed in all further simulations as well (see later). The optimal model values are shown in Table 2 along with the relative change compared with the original model.

It is now essential to check the generality of the obtained set of parameters. Getting much better values for one configuration (in the present case, experiment of [27]) would be of little use if other predictions become worse.

5. Results and discussion

As a consequence, all four turbulent non-premixed ethylene/air flames shown in Table 1 have been computed with the same (optimal) set of parameters. Hereafter, these flames are referred to as Flame I–IV. Flame I has been used to optimize the set of parameters and corresponding comparisons have already been shown in Figs. 3 and 4. Now, Flames II, III and IV are computed with exactly the same physical models, grid size and computational domain.

In Fig. 5 axial profiles of soot volume fraction are shown for Flame II. Experimental data are compared with simulation results both with the original and with the optimal model parameters. Using the original model, the predicted peak value of f_v is nearly 40% lower than experimental data. The optimal parameter set predicts almost exactly the peak soot volume fraction, while a slight shift in peak position towards the burner is again observed.

Table 2
Original and optimal model parameters.

Parameter	Original	Optimal	% Rel. change
T_n [K]	46,100	42,140	−8.59
T_o [K]	26,500	26,218	−1.06
C_0 [$m^{-1}K^{-1}$]	2485	3593	44.0

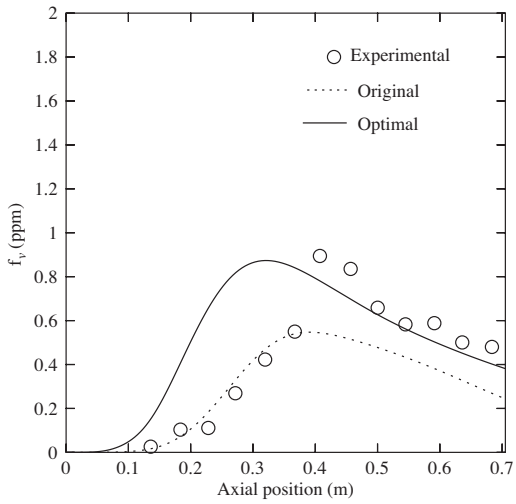

Fig. 5. Profiles of f_v along the flame axis for Flame II [28]. The dotted line represents the original model while the thick continuous line represents the optimal model.

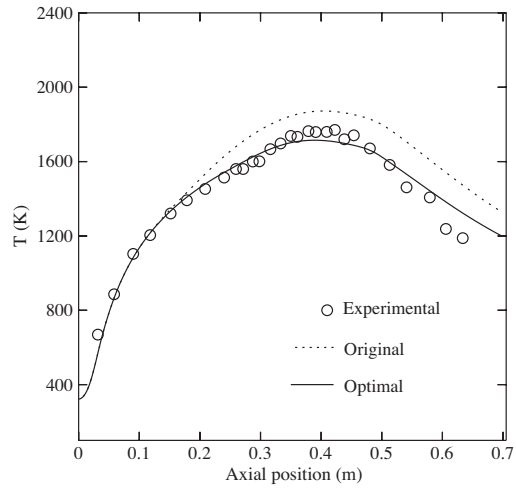

Fig. 6. Profiles of temperature along the flame axis for Flame III [29]. The dotted line represents the original model while the thick continuous line represents the optimal model.

Axial profiles of temperature and soot volume fraction are shown for Flame III in Figs. 6 and 7, respectively. Once again, the agreement concerning temperature is tremendously better with the optimal model compared to the original formulation (Fig. 6). Simultaneously, the peak soot volume fraction is also better predicted with the optimal parameter set, while a shift towards an earlier soot production is again observed (Fig. 7). Note that these measurements are more than 20 years old and might therefore be associated with a higher experimental uncertainty, explaining larger differences between measurements and simulations, independently from the retained numerical model.

Recent experimental studies on turbulent non-premixed ethylene/air flames have been published with varying fuel flow rates (Re = 4000–23, 200) but fixed burner diameter [30]. Due to lack of space, it is not possible to compare all published results here. We therefore retain one single flame configuration at which ethylene is fed into atmospheric air with a velocity of 48.7 m/s, leading to Re = 12,000.

Comparisons of axial profiles for f_v are shown in Fig. 8. Corroborating previous observations, the optimal set of parameters leads to a much better prediction of peak soot volume fraction compared to the original model. Again, this peak is slightly shifted towards fuel inlet compared to

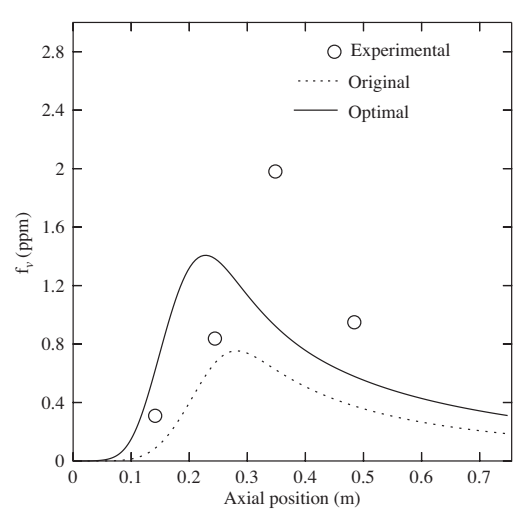

Fig. 7. Profiles of soot volume fraction f_v along the flame axis for Flame III [29]. The dotted line represents the original model while the thick continuous line represents the optimal model.

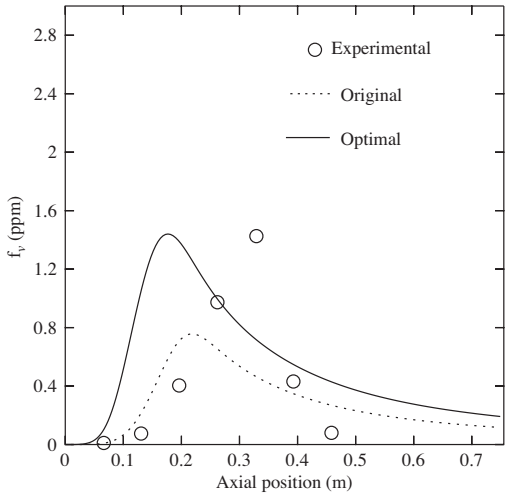

Fig. 8. Profiles of f_v along the flame axis for Flame IV [30]. The dotted line represents the original model while the thick continuous line represents the optimal model.

the original parameter set with both results documenting a faster soot production compared to the experimental measurements.

In order to complement the qualitative comparisons introduced up to now, a quantitative measure of the differences between experimental data and simulation results would be helpful. For this purpose, two different criteria have been computed, as shown in Table 3. The relative norm of the difference is first computed for instance for temperature as the sum over all measurement points $\Sigma|T_{sim} - T_{exp}|/\Sigma T_{exp}$, where the index *sim* denotes simulation results and *exp* is used for experimental data. The relative difference in peak values is computed for instance for temperature as $\Delta H = |T_{sim,peak} - T_{exp,peak}|/T_{exp,peak}$.

It can be clearly observed from Table 3 that the optimal set of parameters leads to a systematic improvement concerning the prediction of peak temperature but also of peak soot volume fraction, an essential criterion for practical applications. Furthermore, the full profiles are also in better agreement for Flames I and III concerning temperature as well as soot volume fraction. Concerning Flames II and IV, the improvement is not as clear, since temperature fields are not known and due to the observed shift towards the burner induced by the optimal set of model parameters. The shape of the profile for f_v is indeed better with the optimal set, but getting rid of the shift should be the next step towards an excellent prediction of soot particles. For this purpose, additional free parameters must be considered in the optimization process.

6. Conclusions

The main objective of the present study is to develop an improved soot prediction model for turbulent non-premixed ethylene/air flames. Optimization relying on Genetic Algorithms coupled with evaluations of the solution by CFD coupled with DQMOM have been carried out for this purpose. The two objective functions of the optimization problem are the absolute differences concerning temperature and soot volume fraction as obtained by CFD and by published experiments. A sensitivity analysis revealed that the three main parameters describing the evolution of the soot particles are the activation temperature of nucleation T_n and of oxidation T_o, together with radiation constant C_0. They have therefore been chosen as parameters of the optimization. The optimal parameter set identified by the procedure leads indeed to a considerable improvement for the target flame [27].

The generality of the optimal set of parameters has then been further tested by comparison with three additional flames, investigated experimentally in the literature. The results show a systematic improvement concerning the most important value for practical applications, i.e., the peak soot volume fraction. The temperature fields are also always improved. Nevertheless, even if the form of the axial profile for soot volume fraction also

Table 3
Quantitative comparison of the results obtained for temperature T and soot volume fracion f_v by comparison with published experimental data. Here, Norm denotes a comparison of the full profile while ΔH is the relative difference in peak values.

Flame	Profile	Norm %		ΔH%	
		Original	Optimal	Original	Optimal
I	T	8.9	3.8	10.7	3.1
	f_v	35.3	32.4	41.9	6.1
II	f_v	32.1	43.7	38.5	2.2
III	T	8.5	2.7	5.8	3.2
	f_v	56.8	56.0	62.0	29.3
IV	f_v	58.8	95.6	47.0	1.0

improves, it is however simultaneously shifted towards the burner. This undesirable shift should be removed in future studies by incorporating further degrees of freedom in the optimization, leading to an even better model. Further comparisons are also necessary.

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